0801IT221055 Faraz Khan

Assignment 1

Installation of Hadoop on a Single Node Cluster

Step 1: Download Hadoop 3.3.6

```
~/…/big_data $ wget -q --show-progress https://downloads.apache.org/hadoop/common/hadoop
-3.3.6/hadoop-3.3.6.tar.gz -P /tmp/
-3.3.6.tar.g  19% 135.64M   737KB/s    eta 2m 27s s          ] 104.64M  4.71MB/s    eta 2m
20s          ] 104.56M  4.70MB/s    eta 2m 20s
```

Running Open in Terminal Cancel

Step 2: Extract Hadoop

```
~/…/big_data $ tar -xzf /tmp/hadoop-3.3.6.tar.gz -C /tmp/
```

Ran terminal command Open in Terminal · Exit code 0

Step 3: Move Hadoop to `/usr/local/hadoop`

```
~/…/big_data $ sudo mv /tmp/hadoop-3.3.6 /usr/local/hadoop
```

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Step 4: Set environment variables in `~/.bashrc`

```
~/…/big_data $ grep -q "HADOOP_HOME" ~/.bashrc && echo "Already set" || cat >
> ~/.bashrc << 'EOF'

# ── Hadoop / Java Environment Variables ──
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
EOF
```

Step 5: Configure Hadoop XML files + update `hadoop-env.sh` for Java 21

```
~/…/big_data $ HADOOP_CONF=/usr/local/hadoop/etc/hadoop

# hadoop-env.sh - point to Java 21
sudo tee $HADOOP_CONF/hadoop-env.sh > /dev/null << 'EOF'
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
EOF

# core-site.xml
sudo tee $HADOOP_CONF/core-site.xml > /dev/null << 'EOF'
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
EOF

# hdfs-site.xml
sudo tee $HADOOP_CONF/hdfs-site.xml > /dev/null << 'EOF'
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>dfs.replication</name>
```

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Step 6: SSH key setup (if not already done)

```
~/…/big_data $ if [ ! -f ~/.ssh/id_rsa ]; then ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa && echo "Key generated"; else echo "SS
H key already exists"; fi
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
echo "SSH setup done"
```

Step 7: Create data directories and format NameNode

```
~/…/big_data $ sudo mkdir -p /usr/local/hadoop/data/namenode /usr/local/hadoop/data/datanode && sudo chown -R $USER /usr/loca
l/hadoop/data && echo "Dirs created and owned"
```

Step 8: Format the NameNode

```
~/…/big_data $ export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64 && export
HADOOP_HOME=/usr/local/hadoop && export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_H
OME/sbin && hdfs namenode -format -force 2>&1 | tail -20
```